

# Victor Gardin

Aerospace Engineering Graduate Student – U.S. Citizen  
Seeking GNC Engineering opportunities in orbital mechanics & mission analysis, Spring/Summer 2027.

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## Education

**Master of Science in Aerospace Engineering** 2025–2027  
*Georgia Institute of Technology – Metz, France & Atlanta, USA*

Relevant coursework: Optimal Guidance & Control, Orbital Mechanics (ongoing), Machine Learning for Robotics

**Arts et Métiers Institute of Technology** 2023–2026  
*Lille, France*

Combined BSc & MSc in Mechanical and Industrial Engineering, dual-degree with Georgia Tech.

Coursework: Solid Mechanics, Fluid Dynamics, Power Electronics, CAD. Admitted via a 2-year Advanced Physics & Technology Preparatory Class (Versailles) to this top-10 French engineering school.

## Research & Work Experience

**Mission Analysis & GNC Engineering Intern** Jan–Jun 2026  
*Agena Space – Bordeaux, France*

Developed a deterministic mission-analysis tool solving Lambert’s problem within an iteratively refined grid search, generating a Pareto front ( $\Delta V$  vs. transfer time) across simultaneous altitude, eccentricity, and inclination corrections; Pareto-dominated the existing sequential-maneuver baseline.

Propagated orbital dynamics via RK4 integration ( $\sim 10^{-3}$  m precision) including J2, solar radiation pressure, and atmospheric drag perturbations; implemented a Kalman filter estimating spacecraft inertial position and attitude state.

Validated GNC robustness on a 50-run Monte Carlo rendezvous/inspection campaign: impulsive-thrust mode achieved zero failures (mean  $\Delta V$  41.9 m/s, sub-meter final dispersion); root-caused a continuous-thrust attitude PID lag error to 5 software defects, which were corrected.

Ported GNC algorithms from Python to C, cutting runtime  $8.2\times$  ( $1.044$  ms  $\rightarrow$   $127.1$   $\mu$ s) with matching outputs; integrated onto a hardware-in-the-loop Flatsat testbed (Raspberry Pi 4, then the OBS-VSP-v1.0 flight computer).

**Graduate Research Assistant** Aug 2026–Present  
*Aerospace Systems Design Laboratory (ASDL), Georgia Tech*

Contributing to a Space Campaign Technologies Assessment evaluating architectures and technologies for sustained lunar surface base operations and logistics.

Building an NLP/LLM-based knowledge-graph pipeline to structure engineering technical documents, improving retrieval speed and traceability for design teams.

**Engineering Intern – System Design & Architecture** Jun–Aug 2025  
*Agena Space – Bordeaux, France*

Developed a Satellite Propulsion Calculator (Python, Java/HTML) computing optimized propulsion configurations from mission parameters.

## Selected Projects

**AI for Engineering Data Retrieval** (with Gulfstream), Georgia Tech – Metz, Fall 2025: Contributed to an LLM-based knowledge-graph system helping engineers search and interpret technical data.

**Competition Propeller Design**, ONERA – Lille, Sep 2024–Feb 2025: Designed and manufactured a propeller focused on noise reduction and energy efficiency.

## Skills

**Programming:** Python, C, C++ (coursework/personal projects), MATLAB/Simulink, Git, LaTeX

**GNC & Mission Design:** Orbital propagation & perturbation modeling (J2, drag, SRP), Lambert-problem trajectory optimization, Kalman filtering, Monte Carlo validation, HIL integration

**Engineering Software:** CATIA V5, SolidWorks, ANSYS

**Languages:** French (Native), English (TOEIC 930/990), Spanish, German